Chapter 8
Thyristors

The thyristor was the dominating switching device in power electronics for a long time. It was described already in 1956 [Mol56] and introduced to the market in the early 60s [Gen64]. The acronym SCR (Silicon Controlled Rectifier) was primarily used for a thyristor in early publications and is still occasionally in use today. In its basic structure, a thyristor can be fabricated without very fine structures and with low-cost photolithography equipment. The thyristor is still widely used in applications with low switching frequencies, such as controlled input rectifiers which are applied at the grid frequency of 50 or 60 Hz. A further actual application field of the thyristor is the power range that cannot be reached with other power devices - the range of very high blocking voltages and very high currents. For high voltage DC power transmission, thyristors with 8 kV blocking voltage and more than 5.6 kA rated current have been introduced in 2008 as a single device in the size of a 6-inch wafer [Prz09].

8.1 Structure and Mode of Function

Figure 8.1 shows the structure of a thyristor in a simplified drawing. The device consists of four layers forming three pn-junctions. The p-doped anode layer is located at the bottom, followed by the n-base, the p-base and finally the n^+-doped cathode layer.

The three pn-junctions formed by the four alternately doped layers are marked by diode symbols J_1, J_2 and J_3 in Fig. 8.1. If a voltage is applied in the forward blocking direction, the junctions J_1 and J_3 are biased in forward direction and the junction J_2 is biased in reverse direction as long as the device is in the forward blocking state. Thus, across J_2 a space-charge region with a high electric field will build up (Fig. 8.1c). It penetrates widely into the weakly doped n^--layer.

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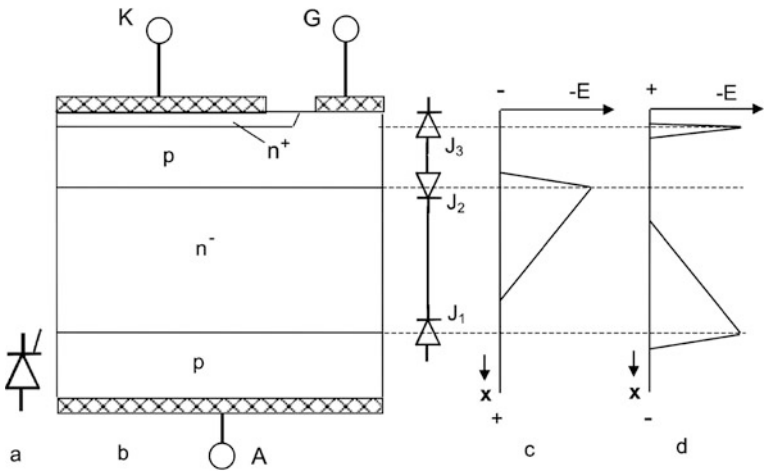


Fig. 8.1 Thyristor. **a** Symbol **b** pn-structure **c** shape of the electric field in the forward blocking mode **d** shape of the electric field in reverse blocking mode

If a voltage is applied in reverse blocking direction of the thyristor, the junction J_2 is forward biased; J_1 and J_3 are biased in reverse direction. Because of the high doping on both sides of the junction J_3, the avalanche breakdown voltage of it is usually relatively low (≈ 20 V). The main part of the applied voltage is taken by the junction J_1, the shape of the electric field is shown in Fig. 8.1d. Since the same weakly doped n^--layer takes the electric field and since the upper and the lower p-layer are usually fabricated simultaneously from both sides in a single diffusion step, the blocking capability of the thyristor is nearly the same for both directions (if the npn-transistor is shorted, see Sect. 8.4). The thyristor is a symmetrically blocking device.

The thyristor can be divided in two partial transistors, a pnp-transistor and an npn-transistor with the common-base-circuit current gains α_1 and α_2, respectively (Fig. 8.2).

Then we obtain for the collector current I_{C1} of the pnp partial transistor according to Eq. (7.1)

$$I_{C1} = \alpha_1 \cdot I_{E1} + I_{p0} = \alpha_1 \cdot I_A + I_{p0}, \tag{8.1}$$

where I_{p0} is the diffusion leakage current from the middle weakly doped n^--layer. In the same way we obtain for the npn partial transistor

$$I_{C2} = \alpha_2 \cdot I_{E2} + I_{n0} = \alpha_2 \cdot I_K + I_{n0} \tag{8.2}$$

with I_{n0} as the diffusion leakage current in the p-base. The anode current I_A is the sum of both partial currents I_{C1} and I_{C2}:

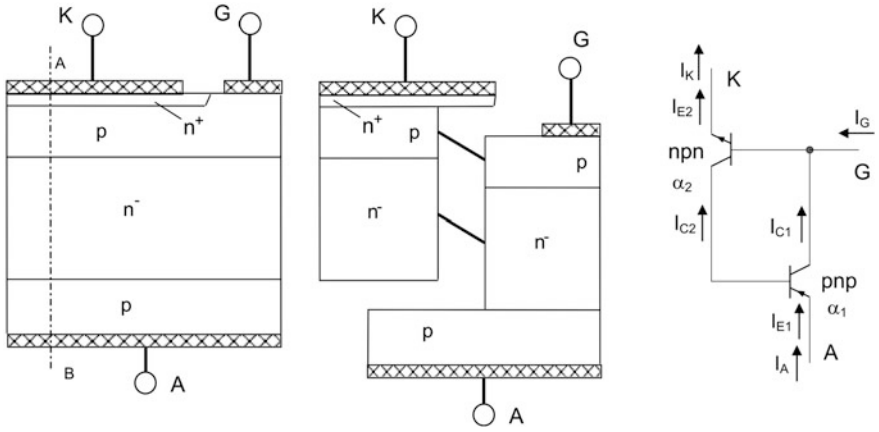


Fig. 8.2 Partition of the thyristor in two partial transistors, and equivalent circuit

$$I_A = I_{C1} + I_{C2} = \alpha_1 \cdot I_A + \alpha_2 \cdot I_K + I_{p0} + I_{n0} \tag{8.3}$$

From the balance of currents flowing into the device and out of the device it holds additionally

$$I_K = I_A + I_G \tag{8.4}$$

Equation (8.4) inserted into Eq. (8.3) leads to

$$I_A = \alpha_1 \cdot I_A + \alpha_2 \cdot I_A + \alpha_2 \cdot I_G + I_{p0} + I_{n0} \tag{8.5}$$

Equation (8.5) resolved for I_A results in an expression for the anode current

$$I_A = \frac{\alpha_2 \cdot I_G + I_{p0} + I_{n0}}{1 - (\alpha_1 + \alpha_2)} \tag{8.6}$$

which can be used as long avalanche multiplication can be neglected. From Eq. (8.6) one can see: I_A rises to infinity, when the denominator in Eq. (8.6) approaches zero. The current gains α_1 and α_2 are in turn dependent on the current. For very low currents they are close to zero, and they increase with current as shown for the bipolar transistor in Fig. 7.7. The trigger condition of the thyristor is therefore

$$\alpha_1 + \alpha_2 \geq 1 \tag{8.7}$$

If the triggering condition is fulfilled, the anode current has the tendency to increase infinitely, this is valid even if $I_G = 0$ in Eq. (8.6). The thyristor is in the forward conduction mode. In this mode, there is an internal positive feedback loop that is

established by the current amplification of the two partial transistors. Both transistors are in the saturation mode, this leads to a low forward voltage drop that is comparable to the voltage drop across a forward biased diode.

Equivalent to Eq. (8.7) is the condition $\beta_1 \beta_2 \geq 1$. At low current, both α and β grow with increasing current, see Fig. 7.7. Especially, said conditions are also fulfilled if the small-signal current gains $\alpha' = \Delta I_C/\Delta I_E$ or $\beta' = \Delta I_C/\Delta I_B$ are used [Ger79], which are larger than α and β at low current. In nowadays fabricated power thyristors, α_2 and β_2 are determined by the cathode emitter shorts and are zero at low gate current, and therefore the trigger function is mainly adjusted by the cathode shorts. Fore more details see Sect. 8.4.

An exemplary diffusion profile of a thyristor along the line A–B in Fig. 8.2 is shown in Fig. 8.3. The fabrication of a thyristor starts with a weakly doped n^--wafer. Usually, both p-layers are created simultaneously by diffusion: Predeposition with an acceptor dopant, e.g. aluminum, at both wafer surfaces and a subsequent high-temperature drive-in step. To create the deep pn-junctions J_1 and J_3, which are typical for high-voltage thyristors, aluminum is a suitable dopant because of its relatively fast diffusion in silicon. To adjust the required doping concentrations at the n^+p-junction J_3 and near the anode contact, additional p-diffusions are applied. Thus, the final doping profile in the p-layers can be approximated by superposition of several Gauss-type profiles. The junctions J_1 and J_2 both exhibit a very shallow gradient of the diffusion profile at the p-side: This facilitates the fabrication of a junction termination structure with beveled edges as shown in Figs. 4.17 and 4.18. The base of the pnp-transistor with thickness w_B and doping concentration N_D determines the blocking capability of the thyristor in both directions.

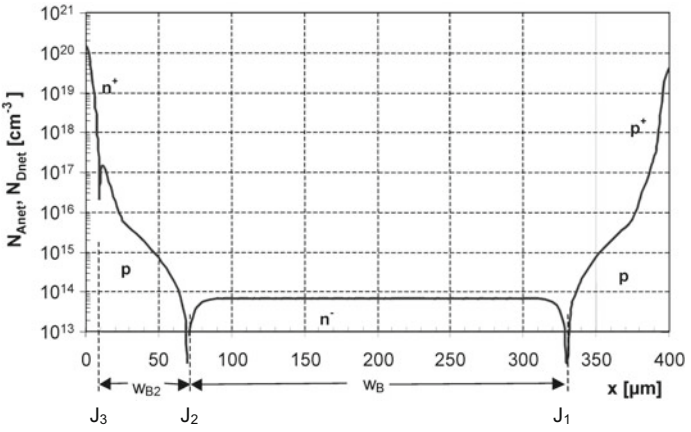


Fig. 8.3 Diffusion profile of a thyristor designed for 1600 V

8.2 I–V Characteristic of the Thyristor

Due to the symmetrical junctions J1 and J2, the blocking behavior of the thyristor is symmetrically in both directions. In forward direction two branches of the I-V characteristic exist: the forward blocking mode and the forward conduction mode. A simplified drawing of the I-V characteristic is shown in Fig. 8.4. In the forward blocking mode, the maximally allowed voltage V_{DRM} is defined at a leakage current I_{DD}. In reverse direction, the maximal voltage V_{RRM} is specified at a maximal allowed current $I_{RD,\ max}$.

The data sheet values for V_{DRM}, V_{RRM} etc. may differ considerably from the values measured for a real device, as mentioned already for the I-V characteristic of diodes. In reverse direction the blocking capability is limited by $V_{R(BD)}$. In forward direction the blocking capability is denoted by the breakover voltage V_{BO}. At a voltage higher than V_{BO} the device is triggered and switches to the forward conduction mode. This mode of triggering, breakover triggering, is usually avoided in power thyristors. Usually the thyristor is fired by the gate. At break-over triggering especially in large-area thyristors, the device may be locally overstressed by uncontrolled local current concentration, and destruction is possible.

In forward conduction mode, the voltage drop V_T is defined at a specified current I_T. The branch of the I-V characteristic for high current is similar to the forward characteristic of a power diode. The middle layer is flooded with free carriers, and similar current densities as in a power diode are possible. Again, as already discussed for the I-V characteristic of power diodes, the data sheet value V_{Tmax}, the maximally allowed forward voltage drop, is higher than the real value V_T of a device, since there is some unavoidable variation in the electrical characteristics of devices. Therefore, the manufacturer usually specifies values with a certain safety margin.

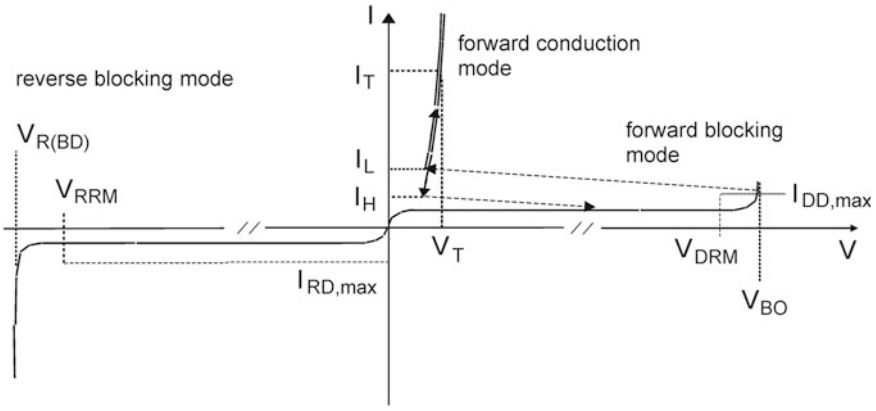


Fig. 8.4 Simplified I-V characteristic of the thyristor and some important thyristor parameters

Dedicated parameters of the forward characteristic are further:

The latching current I_L: The minimal anode current which must flow at the end of a 10 µs trigger pulse to safely switch the thyristor in conduction mode and to maintain the conduction mode safely when the gate signal returns to zero.

The holding current I_H: The minimal anode current necessary to maintain the thyristor in conduction mode without gate current and which ensures, that the conducting thyristor will not extinguish. A decrease of current below I_H can lead to a turn-off of the thyristor.

Since the device is not completely flooded with carriers at the initial phase of the turn-on process, $I_L > I_H$ always applies. The latching current is typically twice as large as the holding current.

8.3 Blocking Behavior of the Thyristor

Avalanche breakdown as limit for the blocking capability is already known from the paragraphs on power diodes and transistors. For a thyristors, there is a second limit of the blocking capability, the punch through effect: The space-charge region, which spreads across the n^--layer with increasing blocking voltage, may arrive at the adjacent layer of opposite doping.[1] Holes will be accelerated in the electric field and the blocking capability is no longer given.

For simplification a triangular electric-field shape across the n^--layer (Fig. 8.1c or d) is assumed in the following discussion. Furthermore, the penetration of the space-charge region into the p-layer of the blocking pn-junction shall be neglected. The avalanche breakdown voltage and its dependency on the background doping were already calculated in Sect. 3.3. It is given by Eq. (3.84) for of the triangular electric field shape. This dependency is drawn in Fig. 8.5 as line (1). It is equal to the dependency in Fig. 3.17, lower part. Additionally, the width of the space-charge region is given by Eq. (3.58). Solving Eq. (3.58) for the voltage and neglecting the small V_{bi} leads to

$$V_{PT} = \frac{1}{2}\frac{q \cdot N_D}{\varepsilon} w_B^2 \tag{8.8}$$

In this equation, the voltage was set to $V_r = V_{PT}$, the voltage at which the space-charge region reaches the region of opposite doping at the position $w = w_B$. For w_B = 250 and 450 µm, the calculated values of V_{PT} are drawn in Fig. 8.5 as line (2) and (3), respectively.

The optimal design parameters for the base width w_B and its doping concentration N_D for a thyristor with a blocking voltage somewhat above 1600 V can be

[1] Already for diodes, the term "punch-through" was used, however in a different meaning of the word. When in diodes the space-charge region extends through the entire n^--layer and penetrates into an n^+-layer, the blocking capability can further increase. See Sect. 5.3.

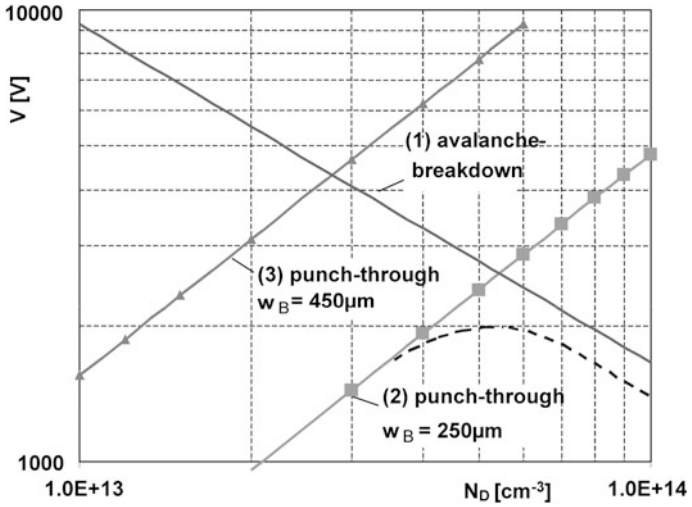


Fig. 8.5 Blocking capability of a thyristor: Avalanche breakdown voltage as a function of N_D, punch-through voltage for two different widths of the n^--layer

estimated by considering the intersection point of lines (1) and (2) in Fig. 8.5. If the doping concentration is reduced below the concentration at the intersection point, the avalanche breakdown voltage will increase, but the space-charge region will reach the opposite p-layer at a voltage lower than the avalanche breakdown voltage and punch-through will limit the blocking capability.

In the following shall be investigated how close one can approach the limits given by avalanche breakdown and punch through. In reverse direction, we neglect the small voltage at J_3 (which is usually shorted, see Sect. 8.4). The behavior at the blocking junction J_1 is equivalent to that of a bipolar pnp-transistor in open base configuration [Her65]. According to Eq. (7.11), the blocking capability of this junction is lower than for a pn-diode. Avalanche breakdown sets on already if $M_p\ \alpha_1 = 1$ holds:

$$M_p = \frac{1}{\alpha_1}. \tag{8.9}$$

Only for $\alpha_1 = 0$, the avalanche breakdown voltage of the pn-junction reaches the value derived for diodes. Since $M_p \ll M_n$ applies (see Fig. 3.15), the effect is not as strong as in an npn-transistor. The onset of avalanche breakdown is reduced to lower voltage, as shown in Fig. 8.5 by the dotted line, the reduction of the breakdown voltage is most pronounced close to the intersection point of lines (1) and (2).

In forward direction the blocking junction is J_2. Its blocking capability is denoted by the breakover voltage V_{BO}. We can use the trigger condition (8.6), set $I_G = 0$ and

take into account the multiplication factors for the hole current in the pnp-transistor and for the electrons in the npn-transistor. The breakover voltage will be reached for

$$I_A = \frac{M_{SC} \cdot I_{SC} + M_p I_{p0} + M_n I_{n0}}{1 - \left(M_p \cdot \alpha_1 + M_n \cdot \alpha_2\right)}. \tag{8.10}$$

For α_1 and α_2 in (8.10), the small signal current gains must be used. The breakover voltage will be reached for $M_p \cdot \alpha_1 + M_n \cdot \alpha_2 = 1$. Since it holds $M_n \gg M_p$, the forward breakover voltage will be very sensitive to α_2. Only for $\alpha_2 = 0$ the blocking capability will be the same as in reverse direction.

Both current gains are dependent on temperature and increase with temperature for low currents. To ensure a blocking capability of the thyristor at higher temperatures, α_2 must be reduced for low currents; since symmetrical blocking capability is required it must be zero for low current. This can be achieved by the implementation of emitter shorts.

8.4 The Function of Emitter Shorts

The current gain of a transistor depends not only on current, but also on temperature. At low temperature the current gain is low; it increases with temperature (Chapter 7, Fig. 7.7). This has the consequence that the breakover condition in open base configuration, Eq. (8.10), will be reached at a lower voltage when the temperature is increased. For the thyristor, the breakover voltage V_{BO} will decrease strongly. This behavior is shown in Fig. 8.6 by the dotted line for a thyristor without emitter shorts.

By the implementation of emitter shorts on the cathode side (Fig. 8.7), a shunt parallel to the pn-junction between the base and the emitter of the npn-partial transistor is created [Chu70, Chk05, Ger79]. The current coming from the pnp-transistor flowing into the base of the npn-transistor is conducted via this shunt

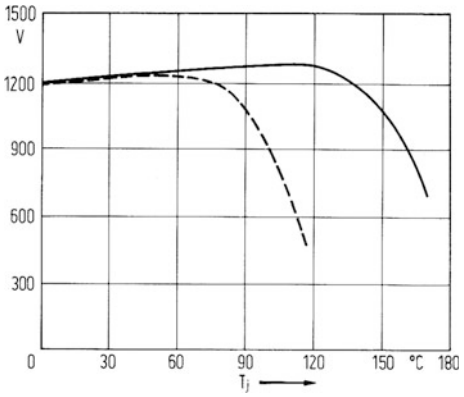


Fig. 8.6 Temperature dependency of the breakover voltage V_{BO}. Dotted line: Without emitter shorts. Full line: With emitter shorts. Figure taken from [Ger79]

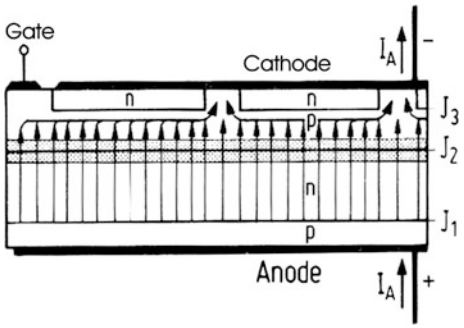


Fig. 8.7 Arrangement of emitter shorts at the cathode side of a thyristor. Figure according to [Ger79]

to the cathode contact. The shunt resistance is determined by the lateral distances between the shunts and the doping concentration of the p-base. If the current is high enough, the voltage drop across the short gets sufficiently high and a perceptible current gain of the npn partial transistor arises.

The cathode emitter shorts determine the effective α_2 and thus widely the dynamic and static characteristics of the thyristor. The dependency of the forward blocking capability on temperature for a thyristor with emitter shorts is drawn in Fig. 8.6 with the full line. With appropriate design of the emitter shorts, the thyristor will have the same blocking capability in forward and in reverse direction even at increased temperature.

Even if a thyristor is provided with emitter shorts, its blocking capability remains sensitive to temperature variations because of the temperature dependency of the current gain of the partial transistors. In most thyristors the maximal allowed operating temperature is restricted to 125 °C, in some special thyristors a little bit above. At increased temperature, thyristors show a significantly increased leakage current compared to diodes.

8.5 Modes to Trigger a Thyristor

A thyristor can be triggered

1. By a *gate current* I_G. This is the most common mode to trigger a thyristor. For technical applications, the following characteristics are given:
   I_{GT}, V_{GT}: Minimal current and minimal voltage that must be provided by a gate unit for securely triggering of the thyristor.
   I_{GD}, V_{GD}: Maximal current and maximal voltage at the gate, at which a thyristor will surely not trigger. To avoid unwanted triggering by disturbing signals, which could occur for example by electromagnetic crosstalk between cables and drive units, these thyristor parameters are very important.

2. By *exceeding the breakover voltage.* In usual power thyristors, this trigger mode is strictly avoided. However, special modifications of thyristor structures, e.g. SIDACs or SIDACtors [SID97], use breakover triggering to act as protection devices against too high voltages. They are connected in parallel to a device or an integrated circuit for protection. They trigger at a voltage higher than V_{BO} and protect other parts of the circuit from overvoltage. Their voltage range is limited to small and medium voltages in the range of some 10 V up to some 100 V. Further, pnpn-thyristor structures are used as electrostatic discharge (ESD) protection structures in integrated circuits, where the same principle of breakover triggering is applied.
3. By a *voltage pulse with a slope dv/dt above the critical* dv/dt_{cr} in forward direction. If such a voltage pulse occurs, the junction capacity of the pn-junction J_2 is charged. If the slope dv/dt is high enough, the generated displacement current may be sufficient to trigger the thyristor. *dv/dt* triggering is an unwanted triggering event. For the application of a thyristor, a maximal allowed dv/dt_{cr} is defined.
4. By a *light pulse* with photons that penetrate into the space-charge region across the junction J_2 [Chk05, Sil75, Sil76]. If the energy of the arriving photons is high enough, electrons transit from the valence band up to the conduction band. The generated electron-hole pairs are separated in the electric field immediately, the electrons flow to the anode, the holes to the cathode. The generated current has the same effect as a current supplied by the gate. If the light power is high enough, the triggering condition Eq. (8.7) can be fulfilled. Light triggering is preferably used in case of series connection of thyristors. This is the case, for example, in applications of high voltage direct current transmission (HVDC), where total voltages up to several 100 kV have to be controlled. The possibility to trigger a thyristor via a glass fiber cable is of high advantage because of the given electrical insulation with this type of signal cable.

The cathode emitter shorts (Sect. 8.4) determine widely the triggering of a thyristor [Sil75]. For activating the npn-transistor, the voltage drop below the emitter to the next emitter short $V = R_{p\text{-}base} \cdot I_G$ must approach the built-in voltage V_{bi} of the n^+p-junction between emitter and p-base, which is in the order of 0.7 V at 300 K. Since V_{bi} will decrease with temperature and $R_{p\text{-}base}$ will increase due to the reduced mobility of holes at elevated temperature, the trigger condition will be much earlier fulfilled at a high operation temperature of 125 °C.

Emitter shorts reduce the trigger sensitivity, increase the trigger current and increase the critical voltage slope dv/dt_{cr} [Sil75]. While the thyristor is most sensitive to dv/dt_{cr} is at 125 °C, it must also be ensured that the necessary trigger current I_{GT} at low temperature – room temperature or down to −40 °C – becomes not too high. The necessary compromise is especially difficult for a light triggered thyristor, where a low trigger power is required in the face of losses e.g. in the glass fiber cables, and still a sufficiently high dv/dt_{cr} must be maintained [Sil76].

8.6 Trigger Front Spreading

With injection of a gate current into a thyristor only the region close to the gate is switched into the conduction mode. The width of the conducting region at the first instant of the turn-on phase is only of the order of some fractions of millimeters. The situation immediately after triggering is illustrated in Fig. 8.8.

The triggered region spreads over the cathode area with a velocity v_z in the range of 50 – 100 µm/µs or 50 – 100 m/s, which is very small for electronic effects. Since there will be only a small lateral voltage drop in lateral direction, the spreading of the initial region of high carrier density is slow. It will take a time of typically 100 – 200 µs until the front of the primary carrier plasma in a thyristor spreads across a length of 1 cm. This slow spreading of the triggered region is a severe limitation for some thyristor applications. It limits the allowed current slope *di/dt* for the anode current.

The trigger front spreading velocity is approximately proportional to the square root of the current density

$$v_z \sim \sqrt{j} \tag{8.11}$$

Furthermore, v_z is reduced by emitter shorts: in the region of shorts it decreases strongly. Emitter shorts reduce v_z to approx. 30 µm/µs for a current density of 100 A/cm^2. Additionally, the carrier lifetime is of influence. If the carrier lifetime is reduced, for example by a gold diffusion, and if, as is necessary for a fast thyristor, strong emitter shorts are implemented, the trigger front spreading velocity v_z may be as low as 10 µm/µs equivalent to 10 m/s – a sprinter at the Olympic Games is running faster!

Thyristors with a higher blocking capability require a larger thickness w_B of the n^--base, v_z decreases with increasing w_B of the thyristor. For a 4.5-kV thyristor with a high carrier lifetime, v_z will be in the range of 20 µm/µs. The dependence v_z on w_B is given by

$$v_z \sim \frac{L_A}{w_B}, \tag{8.12}$$

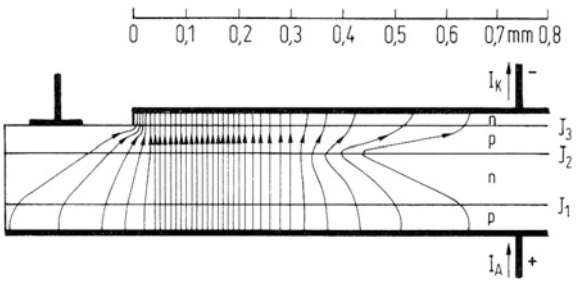


Fig. 8.8 Current distribution in the thyristor immediately after triggering. Figure taken from [Ger79]

where L_A is the ambipolar diffusion length given in Eq. (5.26). A low spreading velocity increases the danger of overstress of the thyristor by an increased local current density in the emitter region close to the gate contact. Therefore, measures to increase the di/dt capability are necessary.

8.7 Follow-up Triggering and Amplifying Gate

Thyristors for applications at line frequency are characterized by a high carrier lifetime and a high ambipolar diffusion length L_A. They are manufactured according to the basic structure described above. Their rated current and *di/dt* capability are of the order of 100 A and 150 A/µs, respectively. These properties are sufficient for applications at line frequency. For applications requiring a higher current capability, it would be possible to increase the gate area, this results in an increase of the initially triggered region. However, this strategy requires a higher gate trigger current I_{GT} and thus a higher effort for the gate drive unit. To avoid this drawback, follow-up triggering was introduced.

First, a pilot thyristor is triggered, which in turn triggers the main thyristor. Thus, the main part of the power necessary for triggering is not provided by the gate drive circuit, but by the main current.

The principle operation of a thyristor structure consisting of a pilot thyristor and a main thyristor is illustrated in Fig. 8.9a. Between the gate and the main cathode K, a small auxiliary cathode K' is arranged. A gate current will trigger the thyristor K'. Its cathode current flows via the resistor R and generates a voltage drop at the resistor. Thus, between K' and K a positive voltage builds up, it generates an electric field in the p-base and a lateral hole current to the n-emitter of the main cathode K, which triggers the main thyristor.

In an advanced configuration the resistor R is integrated into the thyristor (Fig. 8.9b). It is realized by the extension of the n-emitter layer and consists essentially of its lateral resistance in the region reaching from the pilot thyristor to the left border of cathode contact K. This structure was introduced as lateral-field emitter [Ger65] and allows much higher *di/dt* slopes at turn on.

In a thyristor with amplifying gate (AG) [Gen68], shown in Fig. 8.9c, the cathode metal of the auxiliary thyristor K' is connected to the p-base of the main thyristor. After turn-on of the auxiliary thyristor, the current flows across the lateral resistance of the p-base to the main cathode K. The total current of the cathode K' acts as gate current for the main thyristor K. In addition, the overlapping of the cathode contact K' and the p-base forms a cathode short and improves the di/dt capability.

The amplifying gate can be formed in miscellaneous geometrical shapes, for example stripes or other distributed structures spreading across the area of the main cathode. A structure of a large area thyristor is shown in Fig. 8.10. This thyristor is designed for applications of high voltage direct current transmission (HVDC) and has a blocking capability of 8000 V, a rated current of 3570 A (T_{case} = 60 °C), and a di/dt immunity of 300 A/µs.

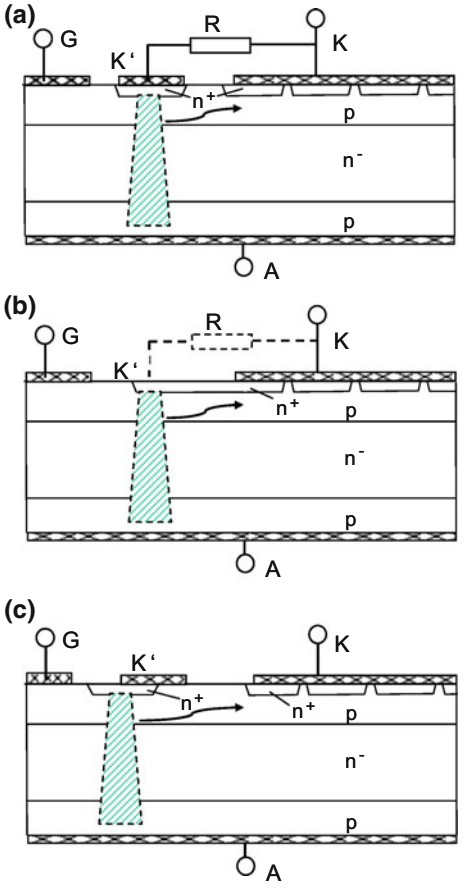


Fig. 8.9 Follow-up triggering of a thyristor. **a** Principle. **b** Lateral-field emitter, integration of the resistor in the n^+ emitter layer. **c** Amplifying gate



Fig. 8.10 Gate structure of a light triggered thyristor, diameter 119 mm. Manufacturer Infineon

The main thyristor area is triggered by a four-stage amplifying gate structure located in the center of the device. The three inner AGs are ring-shaped whereas the fourth AG is distributed about the main cathode. This amplifying gate design ensures triggering of the main thyristor by a wide initial trigger front.

The central structure of such a high-power thyristor can be very complex; especially protection functions can be integrated. Figure 8.11 shows details in the center of the structure in Fig. 8.10 [Nie01, Scu01]. The left edge in Fig. 8.11 corresponds to the chip center in Fig. 8.10. The fourth AG is distributed over the main cathode area to increase the region in which triggering of the main cathode area starts. Direct light-triggering of the thyristor is possible by irradiating the photosensitive area in the center of the device. The typical light power to turn on the thyristor is of the order of 40 mW.

An overvoltage protection is integrated by a breakover diode (BOD) in the center of the structure. Its avalanche current triggers the thyristor via the AG structure when an overvoltage is applied to the device. The voltage level V_{BOD} at which the overvoltage protection function is activated can be adjusted by the distance between the central p region with radius r_{BOD} and the concentric p ring with an inner radius r_p (Fig. 8.11).

The shunt resistance of the weakly doped p^- region below the n^+-emitter of the innermost AG is adjusted such that the *dv/dt* capability of this AG is lower than that of the other AGs and the main cathode area. By this way a reliable *dv/dt* protection function is integrated into the thyristor, because the device is turned on safely by the innermost AG when the anode-to-cathode voltage rises at a rate higher than the rated maximum *dv/dt* value. The resistor R_{23} between the second and the third AGs protects the two innermost AGs from being destroyed when the thyristor is turned on with a high current rate *di/dt*.

A thyristor with a structure in Fig. 8.11 needs high effort in design and fabrication. With the integrated self-protection functions, it includes already elements that support a reliable operation. Integration of these self-protection functions therefore reduces the number of electronic components of the total power electronic system.

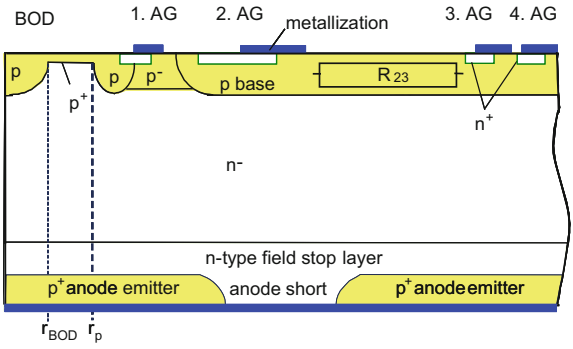


Fig. 8.11 Cross-section of the central amplifying gate structure of a modern high-power thyristor, consisting of four amplifying gates and several protection structures. Figure from [Nie07]

8.8 Thyristor Turn-off and Recovery Time

Only special configurations of the thyristor can be turned off via the gate, these are the Gate Turn Off (GTO) – thyristors which will be treated later. The usual turn-off of a thyristor happens by zero crossing of the anode current, which is given for applications operating in an AC circuit. In case of forward conduction, the base of the thyristor is flooded with free carriers, similar to the internal plasma at forward conduction of a diode. At commutation into the reverse direction, a reverse current will occur, because the stored charge must be removed. This process is similar to the turn-off process in a pin diode. The stored charge must be removed down to a very small rest charge, before a voltage in forward direction can be re-applied to the thyristor. The time which is necessary for this charge removal, and which must be minimally set as hold-off interval to avoid unwanted triggering of the thyristor, is denoted as the recovery time t_q. A thyristor is not able to withstand a forward voltage pulse with the rated blocking voltage or the rated maximum dv/dt_{cr} value until the charge-carrier plasma is almost completely removed from the n^--base.

Figure 8.12 shows the definition of the recovery time. The current slope di_T/dt of the anode current is determined by the external circuit. Equation (5.68) is valid similar as in diodes. The anode current crosses zero, and a reverse current peak occurs due to the removal of the stored charge.

In a typical thyristor first the junction J_3 is depleted of charge carriers when the current is turned-off. However, the blocking capability of this junction is usually only between 10 and 20 V, because of the relatively high doping concentration of the p-base in the range of 10^{17} cm^{-3}, and additionally it is provided with cathode shorts. Thus, the reverse current is increasing until the junction J_1 is depleted of carriers. Then, the thyristor starts to take over the reverse voltage V_R, and a short instant later the reverse recovery current reaches its maximum value.

After having reached the maximum, the reverse current decreases and a voltage peak V_{RM} is generated, similar to the behavior of a diode during the turn-off period.

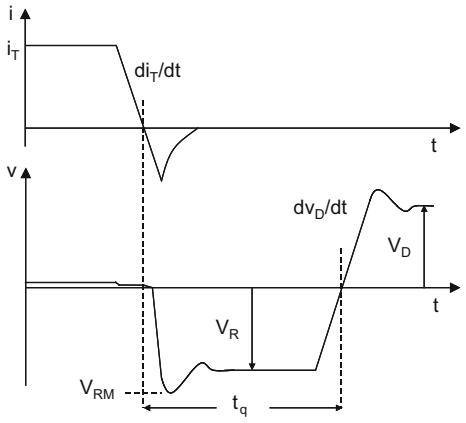


Fig. 8.12 Definition of the recovery time t_q of a thyristor

For thyristors, this period is less critical than for fast diodes. One reason for this is the large width w_B of the n^--layer typically used in thyristors. In most cases, the reverse current decays slowly and a tail current is observed. During this period, charge carriers are still stored in the region close to the junction J_2.

While the thyristor is in the reverse blocking mode, the polarity of the applied voltage changes. A voltage V_D with a defined slope dv_D/dt is applied in forward direction, the thyristor remains in the blocking mode (forward blocking mode) and does not switch into the on-state mode, if $t > t_q$ applies (Fig. 8.12). Otherwise, the control of the circuit is lost. The thyristor must be able to withstand the applied forward voltage, thus, the forward voltage V_D is allowed to be applied only after a certain time interval. This minimum time interval between zero crossing of the current i_T and zero crossing of an applied forward voltage V_D is denoted as turn-off time t_q.

The recovery time of a thyristor is much higher than the switching time of a diode. For the case of charge removal without applying a voltage V_R in reverse direction, and neglecting also V_D, one can estimate according to [Ger79]

$$t_q \approx 10 \cdot \tau, \tag{8.13}$$

where τ is the charge carrier lifetime in the n-base. However, t_q is specified at high operation temperature, whereas τ is typically measured at room temperature. In modern thyristors, t_q depends strongly on the cathode shorts. For turn-off with an applied reverse voltage V_R, Eq. (8.13) can only be used for an estimation of the upper limit value of t_q [Ger79]. If a voltage in reverse direction is applied, only in that part of the base, in which the space-charge region has build up, the stored carriers are removed efficiently by the electric field. The recovery time t_q depends on the application conditions:

- the forward current I_T: t_q increases with increasing forward current;
- the temperature: t_q increases with temperature because τ is increasing with temperature;
- the voltage slope dv/dt. This voltage slope must be smaller than the critical voltage slope dv/dt_{cr} in any case. The closer dv/dt approaches dv/dt_{cr}, the less rest charge is allowed, and the higher is t_q.

With a 100 A 1600 V thyristor, which is in its basic form triggered via a gate in the center, t_q is in the range of 200 µs. With a high-power 8-kV thyristor, t_q is in the range of 550 µs. In large-area thyristors, a forward voltage pulse appearing before t_q may turn on the thyristor somewhere in the main cathode area in an uncontrolled way and – in the worst case – may lead to a destruction of the thyristor. Special structures have been introduced to integrate a t_q protection function into the thyristor [Nie07].

The recovery time limits the maximal frequency range allowed in thyristor applications. With diffusion of gold t_q can be reduced, and also with a higher density of cathode shorts. Fast gold-diffused thyristors reached a t_q down to 10 – 20 µs. Since modern power devices with turn-off capability are meanwhile available,

the interest in fast thyristors has vanished. For the high voltage range > 3 kV, the development of fast thyristors was never successful.

8.9 The Triac

In a triac (triode AC switch) two thyristors are integrated in an anti-parallel configuration in a single device. The triac was introduced early [Gen65]. Figure 8.13 shows the structure.

For a triac, one can no longer distinguish between anode and cathode, therefore the notations "Main Terminal 1" and "Main Terminal 2", MT_1 and MT_2 are conventionally used.

The triac can be triggered via a common gate in both directions. Its I-V characteristic presents in both the first and the third quadrant a conducting branch and a blocking branch. The triac may replace two thyristors in an AC converter, but only within some limits.

The triac is described in more detail in [Ger79]. For the application, the main restriction is due to fact that at zero crossing of the current the triac must block the voltage in reverse direction. However, in conduction mode, the device is flooded with free charge carriers. If the commutation is executed with a too high *di/dt*, still a part of the stored charge remains after zero crossing of the current. If now a voltage with a too high *dv/dt* is applied, an unwanted re-triggering occurs. The device will not transit to the blocking block mode and the possibility to control the converter is lost.

Therefore the allowed current slopes *di/dt* and voltage slopes *dv/dt* are limited drastically for a triac—*di/dt* to some 10 A/µs and *dv/dt* to the order of 100 V/µs. This allows the use of triacs only for applications with comparatively small current and moderate voltage. In fact, in such applications they are often used. An example for triac application is the AC converter for controlling of a medium-power heater.

If currents higher than 50 A have to be controlled, two thyristors in anti-parallel configuration are commonly used instead of a triac.

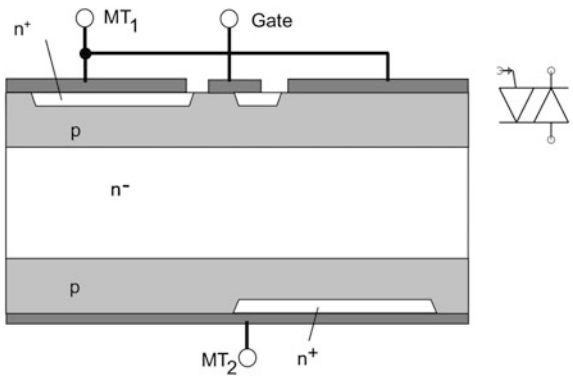


Fig. 8.13 Triac structure and symbol

8.10 The Gate Turn-off Thyristor (GTO)

To provide a thyristor with an active turn-off capability, several special measures are necessary. The Gate-Turn-Off (GTO) thyristor was introduced in the 1980s [Bec80]. In the voltage range above 1400 V it was superior to the bipolar power transistor, the competing device at that time. But with the introduction of the IGBT (see Chap. 10) and the capability to design the IGBT for a high voltage, the GTO thyristor was superseded by the IGBT, since a GTO thyristor requires a high negative gate current for turn-off and the effort for the drive unit is high. The GTO thyristor is nowadays used in the power range that is not reached by the IGBT. Up to 6 kA 6 kV GTO thyristors fabricated from one single 150-mm wafer are available [Nak95]. From the GTO-structure, the new device Gate Commutated Thyristor (GCT) was derived. It exhibits an improved ruggedness and an extended safe operation area.

Equation (8.6) was derived for the trigger condition of a thyristor and describes the dependency of the trigger condition on the current gains of both partial transistors. From this equation, also a turn-off condition can be derived. If in Eq. (8.6) the leakage currents of the partial transistors are neglected, we obtain:

$$I_A = \frac{-\alpha_2 \cdot I_G}{(\alpha_1 + \alpha_2) - 1} \tag{8.14}$$

For turn-off, a negative gate current—I_G is necessary. Similarly to the current gain β for the bipolar transistor, a turn-off gain β_{off} for the turn-off process of a GTO thyristor can be defined:

$$\beta_{off} = \frac{I_A}{-I_G} \tag{8.15}$$

With Eq. (8.14) it follows for the turn-off gain:

$$\beta_{off} = \frac{\alpha_2}{(\alpha_1 + \alpha_2) - 1} \tag{8.16}$$

A high turn-off gain requires on the one hand a high current gain α_2 of the npn partial transistor, on the other hand the denominator $(\alpha_1 + \alpha_2 - 1)$ must be small and ideally approaches to zero. In other words, the sum of the current-gain factors, $\alpha_1 + \alpha_2$, should be only slightly larger than 1. However, this leads to an increased trigger current I_{GT}, an increased latching current I_L, and finally to an increased forward voltage drop V_T of the thyristor. The demand for a high current gain is therefore in contradiction to the requirement for low conduction losses. GTO thyristors show typically a turn-off gain β_{off} between 3 and 5. Thus, to turn off a 3000-A GTO thyristor for example, the drive unit must supply a current of 1000 A.

In fact, Eq. (8.16) is of low value for the design of a GTO thyristor with high turn-off capability. Most important is the homogenous operation of the large number of segments in a high-current GTO thyristor [Shi99].

Further, Eqs. (8.14 - 8.16) hold only if the lateral voltage drop of the turn-off current below the emitter can be neglected [Wol66]. The observance of conditions (Eqs. (8.14 - 8.16)) alone is not sufficient to achieve a gate-turn-off capability of a thyristor. The GTO is thyristor distinguished from the conventional thyristor by its emitter structure, which consists of separated emitter fingers (Fig. 8.14). The width b of the fingers must be small, since the charge carriers below the emitter fingers must be extracted via the gate contact at turn-off. The width b is typically between 100 and 300 µm in modern GTO thyristors. The GTO thyristors consist of a large amount of emitter fingers. They are typically used to control very high currents, therefore a large device area is necessary and it is usual to fabricate a single GTO thyristor from a wafer.

Figure 8.15 shows such a GTO thyristor, fabricated from a 100-mm wafer. The gate contact is formed as a ring located between four inner and four outer rings with emitter fingers. The main reason for this arrangement is to ensure that the distance of the gate contact from the most distant fingers is small enough to allow an efficient extraction of the charge carriers, and that the voltage drop in the gate metallization is not too high.

During turn-off, the holes in the p-base are removed by the negative gate voltage and flow towards the gate contact. The charge carrier plasma which transports the anode current is first removed from the edge of the emitter finger, the residue of the plasma is located in the center of the finger (Fig. 8.16). The hole current must flow laterally underneath the emitter finger. Before the anode current finally is interrupted, a small region in the middle of the emitter fingers, or even of only very few single emitter fingers, takes the total anode current. This is the weak point of the GTO thyristor. To achieve a turn-off capability for high current, it is essential that the resistance of the p-base below the emitter finger is not too high.

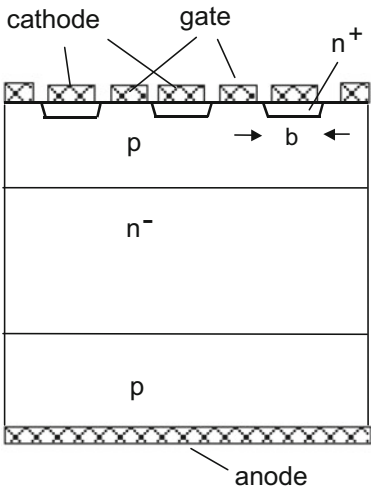


Fig. 8.14 Gate-Turn-Off (GTO) thyristor

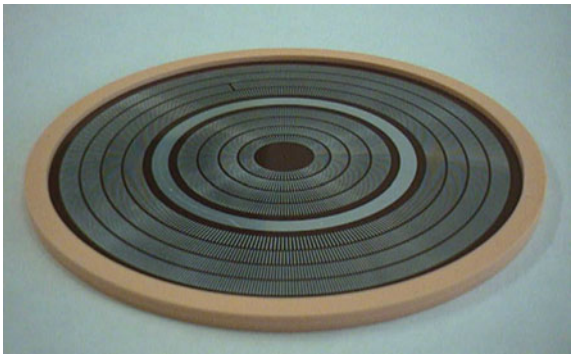


Fig. 8.15 Arrangement of the emitter fingers of a 4.5 kV GTO thyristor fabricated from a 100 mm silicon wafer; final device diameter 82 mm. Figure from Infineon

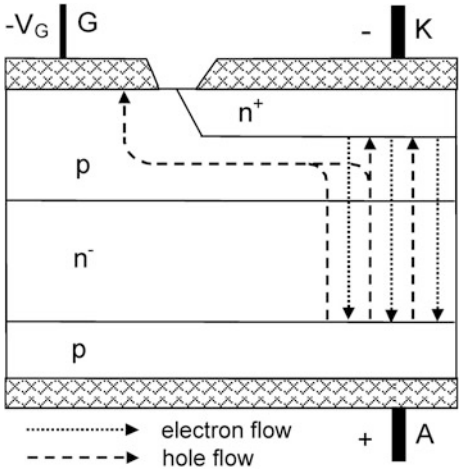


Fig. 8.16 Current flow in a single finger of the GTO thyristor at turn-off

The maximal current I_{Amax} that can be turned off by a GTO thyristor is determined by the breakdown voltage $V_{GK(BD)}$ of the n^+p-junction between the cathode and the gate and by the lateral resistance R_p of the p-base below an emitter finger:

$$I_{\mathrm{Amax}} = \beta_{off} \cdot \frac{V_{GK(BD)}}{R_p} \tag{8.17}$$

where

$$R_p \sim \rho\, b \tag{8.18}$$

and ρ is the specific resistivity of the p-base below the emitter finger. In a GTO thyristor with fingers of a width b = 300 µm, the resistivity ρ below the emitter must be four times smaller than in a conventional thyristor. This requires a sufficiently high p-base doping concentration N_A. Simultaneously, a sufficient blocking voltage $V_{GK(BD)}$ of the n^+p-junction between cathode and gate is required. This

blocking voltage is given by Eq. (3.84). However, the doping concentration that determines the breakdown voltage of the emitter-gate junction in the GTO thyristor is the doping concentration N_A of the p-base. Therefore, N_A must not be too high. Typical doping concentrations are of the order of 10^{17} cm^{-3}, resulting in breakdown voltages $V_{GK(BD)}$ of about 20 - 22 V. The applied gate voltage at turn-off is usually −15 V.

Values > 4 for β_{off} do not increase the turn-off capability significantly. Decisive for the GTO-design is the second term on the right side of Eq. (8.17).

With these measures the plasma can be efficiently extracted from the p-base of the GTO thyristor. However, the plasma of charge carriers still remains in the wide n^--layer. Thus, additional measures are necessary to remove the charge carriers in this region. The first GTO thyristor generation utilized a gold diffusion to adjust a low carrier lifetime in the n^--layer. However, the gold diffusion is very difficult to control with sufficient accuracy (see Sect. 4.9).

An effective improvement was the implementation of shorts at the anode side. The structure of a GTO thyristor with anode shorts is shown in Fig. 8.17. The hole current is extracted via the gate and the injection of electrons from the n^+-emitter is stopped. The electrons in the n-base are removed via the anode shorts by the high positive voltage at the anode side. The injection of the anode emitter is interrupted, and the charge carriers are removed effectively.

A GTO thyristor with anode shorts has no blocking capability in reverse direction. In most applications this is no disadvantage, since an inverse freewheeling diode is connected in parallel to the GTO thyristor in the power circuit. In modern GTO thyristors, anode shorts and charge carrier lifetime control are combined. An implantation of protons or helium nuclei is preferably used for adjustment of the carrier lifetime. The region with high density of recombination centers

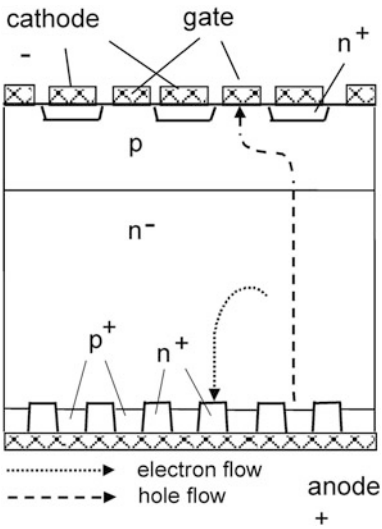


Fig. 8.17 GTO thyristor with emitter shorts at the anode side

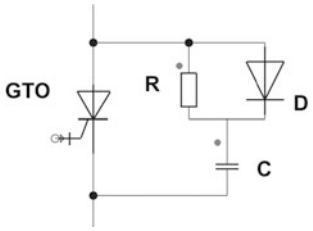


Fig. 8.18 RCD snubber circuit for a GTO thyristor

is located close to the p^+-anode layer, since at this position their effect on the stored charge is most effective.

Despite all these measures, the slope *dv/dt* of the voltage applied to a GTO thyristor at turn-off must be strongly restricted. This is done by an RCD circuit, often denoted as "snubber" (Fig. 8.18). The slope *dv/dt* of the increasing voltage is limited by the capacitor C.

Figure 8.19 finally shows the turn-off process of the GTO thyristor. The negative gate current increases up to the value I_{GRM}, just then the anode current begins to fall. The turn-off delay time t_{gs} is defined as time interval between the moments when the gate current I_G crosses zero and the anode current is dropped to 90% of starting anode current I_{T0}. The anode current then falls steeply within the fall time t_{gf}. During this period, a voltage peak V_{pk} occurs in the waveform of the anode voltage. The value of V_{pk} is determined by the parasitic inductance in the snubber circuit and by the forward recovery voltage peak V_{FRM} of the snubber diode D, the last term typically dominates. Just after V_{pk} the effect of the snubber starts. The slope *dv/dt* of the voltage is then limited by the capacitor C.

In a GTO thyristor, a tail current follows after the interval t_{gf}. This tail current is generated by the extraction of the stored charge in the part of the n-base close to the anode junction. Its duration is of the order of several microseconds, and it generates the main part of the switching losses during the turn-off phase. The implementation of effective anode shorts and the adjustment of the charge carrier lifetime reduce the tail current.

Even if the gate control unit is properly designed, two further factors are of disadvantage for the application of GTO thyristors:

1. The requirement for an RCD snubber. For a high voltage > 3 kV, the capacitor is very voluminous and expensive, especially under the additional constraint of low internal inductance.
2. As already shown, the charge removal below the emitter fingers starts at the finger edges. Before the anode current is decreasing, a narrow region in the centre of the finger remains which carries the total anode current. The larger the device, the more difficult it is to achieve a homogeneous operation of all fingers, and it may happen at the end of the turn-off period that a few of them or even a single finger has to carry the total anode current. This is the weak point of the GTO thyristor, because the respective finger may be destructed under such conditions.

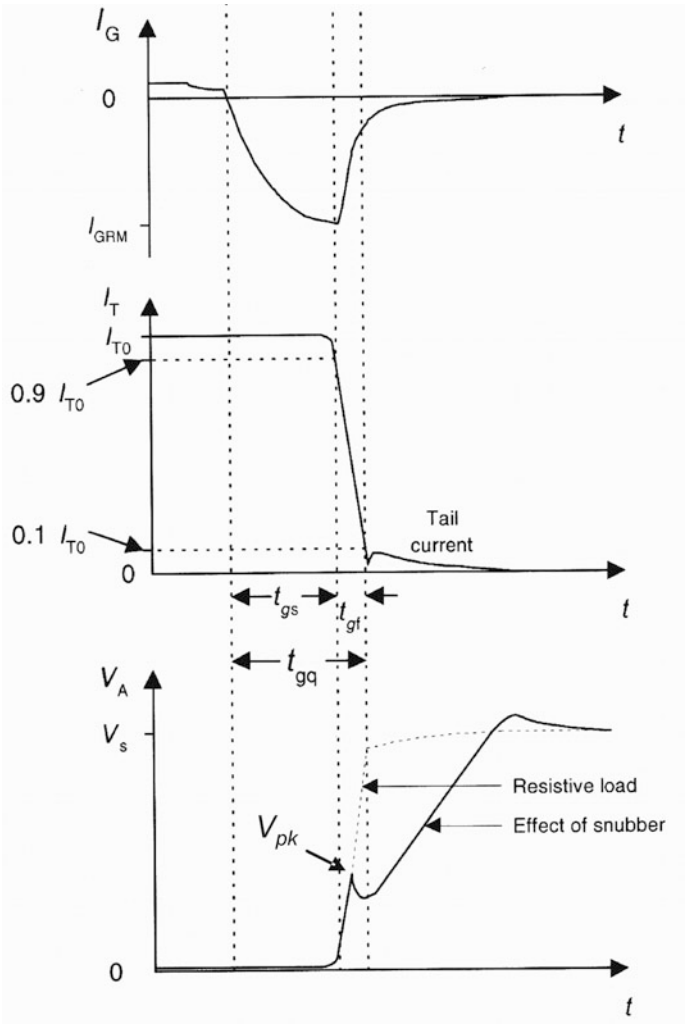


Fig. 8.19 Turn-off characteristic of a GTO thyristor. Figure taken from [Ben99]

8.11 The Gate Commutated Thyristor (GCT)

The operation principle of the GCT [Gru96] is to work with a drive unit that has the capability to transfer the total anode current within a very short time into the gate unit. The GCT is turned off with a turn-off gain $\beta_{off} = 1$.

The GCT consists of the semiconductor device, a gate connection with very low inductance realized by a printed circuit board PCB in coplanar design, and the drive unit assembled with low-inductive capacitors and low-resistive MOSFETs (Fig. 8.20). The drive unit must be capable to deliver the gate current in the range of

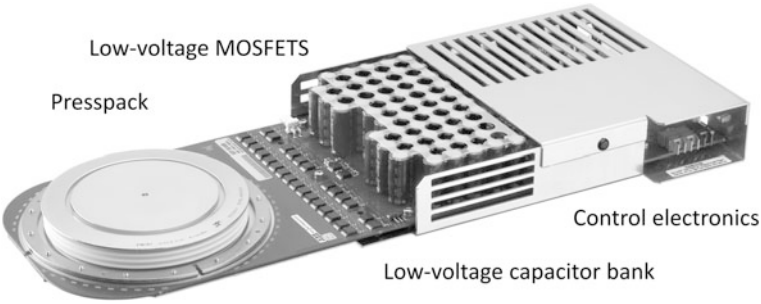


Fig. 8.20 GCT with gate drive unit. The gate wiring consists of wide metal layers on the PCB and is connected in form of a ring to the device. Fig. from [Vem15] © EPE ECCE Europe 2015 – European Conference on Power Electronics and Applications

the anode current within one microsecond. This is a severe challenge; especially the resistance R_G and the parasitic inductance L_G in the gate drive circuit must be extremely low. In the gate drive circuit, the differential equation holds [Lin06]:

$$L_G \cdot \frac{di_G}{dt} + R_G \cdot i_G = V_G \tag{8.19}$$

The solution of this equation is

$$i_G(t) = \frac{V_G}{R}\left(1 - \exp\left(-\frac{R_G}{L_G} \cdot t\right)\right) \tag{8.20}$$

The voltage V_G is limited by the avalanche breakdown voltage of the pn-junction between gate and cathode. The requirement is that the gate current must increase within a time t_{gs} up to the anode current I_A. Thus, the maximally allowed inductance L_G is given by

$$L_G = -\frac{R_G \cdot t_{gs}}{\ln\left(1 - \frac{R_G}{U_G} \cdot I_A\right)} \tag{8.21}$$

The maximally allowed inductance L_G is shown in Fig. 8.21 as a function of the gate resistance R_G for the conditions V_G = 20 V, anode current I_A = 4000 A and allowed t_{gs} = 1 µs. The resistance R_G and the parasitic inductance L_G must be kept extremely small to allow an operation as GCT. For R_G > 5 mOhm, no solution exists for the specified values of V_G, I_A and t_{gs}.

It must be considered that L_G as well as R_G consist not only of the external wiring. The resistance R_{on} of the used MOSFETs, the wiring, the resistance of the connections within the semiconductor housing and the resistance of the gate metallization of the semiconductor die contribute to R_G. The MOSFETs must have a blocking capability slightly higher than 20 V. In this voltage range very low-resistive

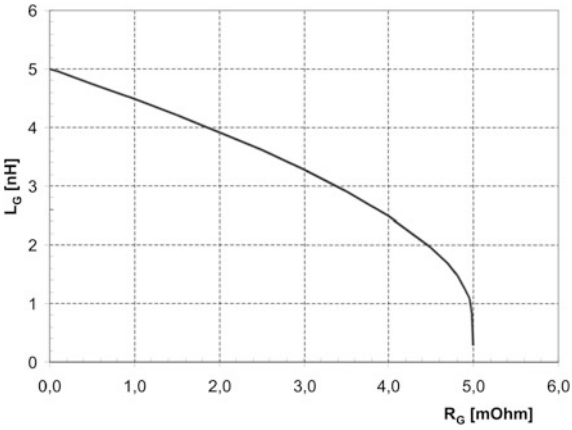


Fig. 8.21 Maximum allowed parasitic inductance L_G in the gate circuit of a 4000 A GCT. Figure according [Lin06]

Si-MOSFETs are available today. In this operation mode the npn partial transistor is turned off abruptly. The positive feedback loop in the thyristor is interrupted. The extraction of the charge carries below the emitter fingers still starts at the edges of the fingers, but the problem of narrow filaments during turn-off is solved to a great extent. The GCT can be operated without RCD-snubber.

There is no requirement for a reverse blocking capability of a GCT. Therefore, it is possible to implement an n-doped buffer layer in front of the anode region and to design the device with a moderate trapezoidal shape of the electric field (PT dimensioning). With this additional buffer layer, the thickness w_B of the n^--layer can be reduced. Consequently, conduction losses and turn-off losses are diminished.

A drive unit which must supply a current as high as the current to be controlled requires high effort without any doubt. But the power, which must be supplied by the driver unit, is not higher than the driver power for a GTO thyristor with the same current turn-off capability. In contrast, it is reduced. In a GCT the negative gate current increases rapidly and the high current flows only during a short time interval. The total charge which is extracted via the gate is even smaller than in a GTO thyristor, since in a GTO still new carriers are injected by the emitter junction J_3 during the increase of the negative gate current in the interval t_{gs} (Fig. 8.19), i.e. the n^+-emitter replenishes part of the charge that must be extracted by the gate driver. In a GCT, only the charge stored before turn-off must be removed. The GCT can be operated with even half of the drive power of a GTO [Lin06]. Although only few modifications of the silicon die are made in a GCT compared to a GTO, the CCT constitutes a significant progress. The main weaknesses of the GTO are solved to a large extend.

The Integrated Gate Commutated Thyristor (IGCT) contains a monolithically integrated diode, see Fig. 8.22. The carrier lifetime adjustment in the diode area is executed with particle irradiation, see Sect. 4.9. There must be no cumbersome compromises in diode performance [Kla97].

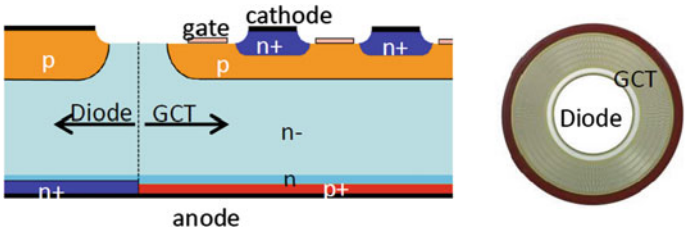


Fig. 8.22 IGCT with integrated Diode. Fig. adapted from [Kla97]

Figure 8.23 shows the IGCT-presspack internal construction. The center of the device (middle, top) is the diode area. It is contacted on bottom with a uniform molybdenum disk (bottom left) and on top side with 2 molybdenum disks for GCT and diode are. The top-side pressure camp, bottom middle, contains the ring-shaped gate with radial interconnections. They realize the low-inductive interconnection to the PCB.

A further progress for the GCT is the corrugated p-base, shown in Fig. 8.24. It is fabricated by an ion implantation through a mask covering mainly the n^+ segment. After the drive, the p-base has two different junction depths with an intermediate region with more shallow penetration depth. [Wik07]. An increased of the maximal current that can be turned-off was found, the increase is a factor of 1.4 at 25 °C and a factor of 1.1 at 125 °C.

A reason is that the position of highest current at turn-off, which is in the middle of the emitter finger (Fig. 8.16), is now different from the position of high electric field when the junction is subject to reverse voltage, as it is the case at turn-off. This



Fig. 8.23 IGCT Presspack internal construction. Reprinted with permission from ABB

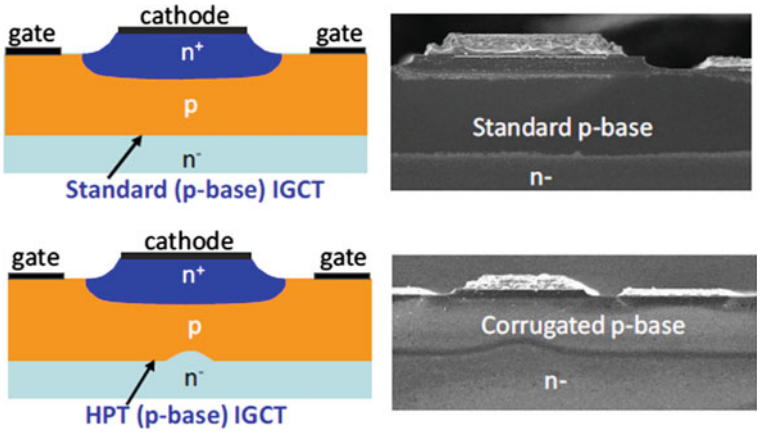


Fig. 8.24 High Power Technology (HPT) GCT with corrugated p-base design compared to a standard uniform p-base junction. Fig. from [Vem15] © EPE ECCE Europe 2015 – European Conference on Power Electronics and Applications

laterally modulated electric field helps to defocus the generated holes from the center of the emitter finger during dynamic avalanche [Wik07]. For further details on dynamic avalanche, see Chap. 12.

The GCT hat the potential of very low conduction losses. It is shown in [Rah13] that for 2.5 kV GCT (91 mm diameter) a forward voltage drop even below 1 V can be achieved at 1500 A, if the intended switching frequency is 125 Hz. This can be used for high-current applications where very high efficiency is required.

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